

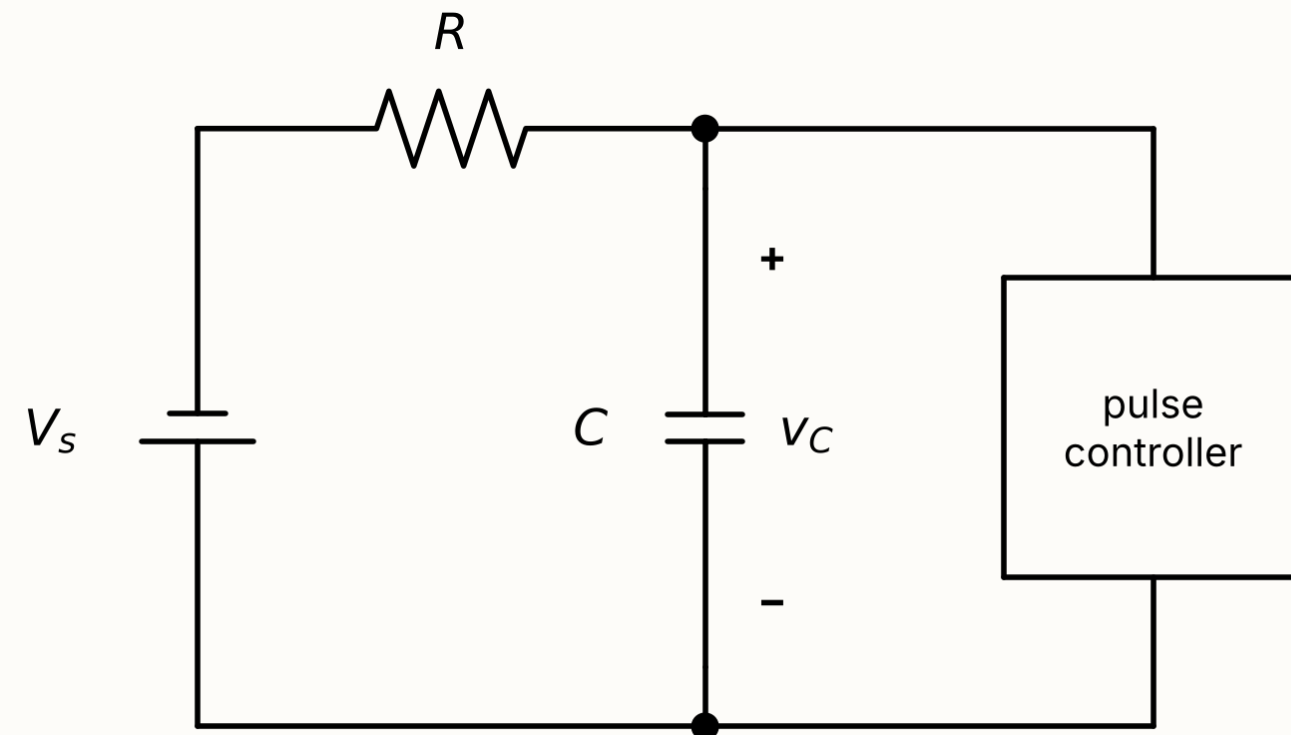
Natural and Step Responses of RL and RC Circuits

Flip a switch, and every current and voltage glides from where it was to where it is going. Three numbers tell you how.

Prepared by Dr. Adel Daaboul

PREDICT FIRST

In Nilsson's model, a pacemaker times each heartbeat with one resistor and one capacitor. Double the resistor: does the heart beat faster or slower?



faster

slower

no change

CHAPTER 7 GOALS

By the end of Chapter 7 you can

1 • Natural response

Find how stored energy drains from an inductor or a capacitor through resistors.

2 • Step response

Find how a dc source drives the current or voltage to a new value.

3 • One method

Find the initial value, the final value and τ , and write the answer in one line.

4 • Check it

Say what can jump at $t = 0$, and check an answer against the circuit.

One inductor or one capacitor: one exponential

$$x(t) = x_f + (x_0 - x_f) e^{-t/\tau}, \quad \tau = \frac{L}{R_{Th}} \text{ or } \tau = R_{Th} C$$

- x is the inductor current or the capacitor voltage: the only quantities guaranteed not to jump.
- x_0 from the circuit before switching; x_f from the dc circuit long after (inductor short, capacitor open).
- R_{Th} is the resistance the element sees after switching, with the independent sources off.
- Natural response: no source after switching, so $x_f = 0$. Step response: a dc source, so $x_f \neq 0$.

THE NATURAL RESPONSE

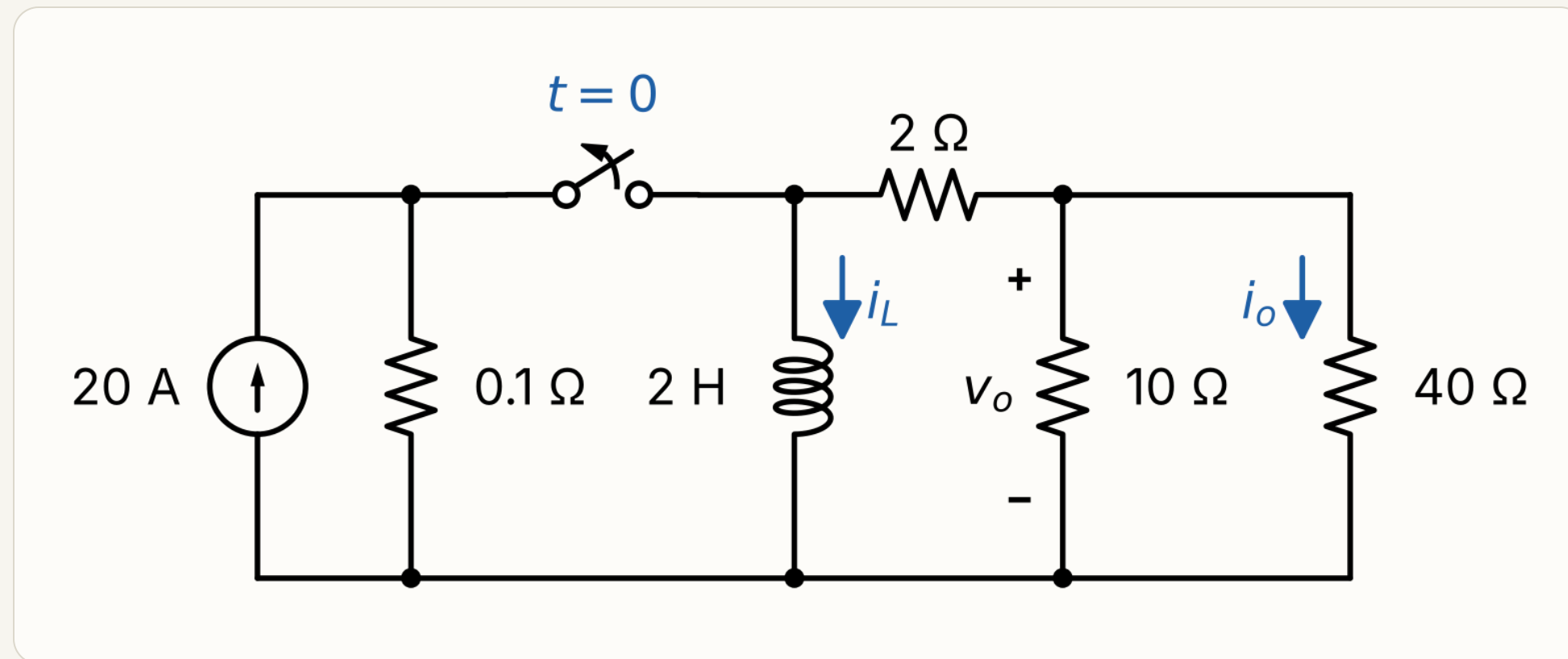
Stored energy drains through the resistors

$$i(t) = I_0 e^{-t/\tau}, \quad \tau = \frac{L}{R} \qquad v(t) = V_0 e^{-t/\tau}, \quad \tau = RC$$

- 1** Find the initial inductor current (or capacitor voltage) from the circuit for $t < 0$.
- 2** Find the equivalent resistance the element sees for $t \geq 0$, and τ .
- 3** Write the exponential.
- 4** Find every other current and voltage from it.

NILSSON EXAMPLE 7.1

The switch opens at $t = 0$ after a long time closed



For $t < 0$ the inductor shorts the $0.1 \, \Omega$ and everything to its right, and carries all 20 A.

$$i_L(0) = 20 \, \text{A}, \quad R_{eq} = 2 + 40 \parallel 10 = 10 \, \Omega, \quad \tau = \frac{2}{10} = 0.2 \, \text{s}, \quad i_L = 20e^{-5t} \, \text{A}$$

NILSSON EXAMPLE 7.1

Resistor currents can jump; the inductor current cannot

The inductor current divides between the 10 Ω and the 40 Ω :

$$i_o = -i_L \frac{10}{10 + 40} = -4e^{-5t} \text{ A}, \quad v_o = 40 i_o = -160e^{-5t} \text{ V}, \quad t \geq 0^+$$

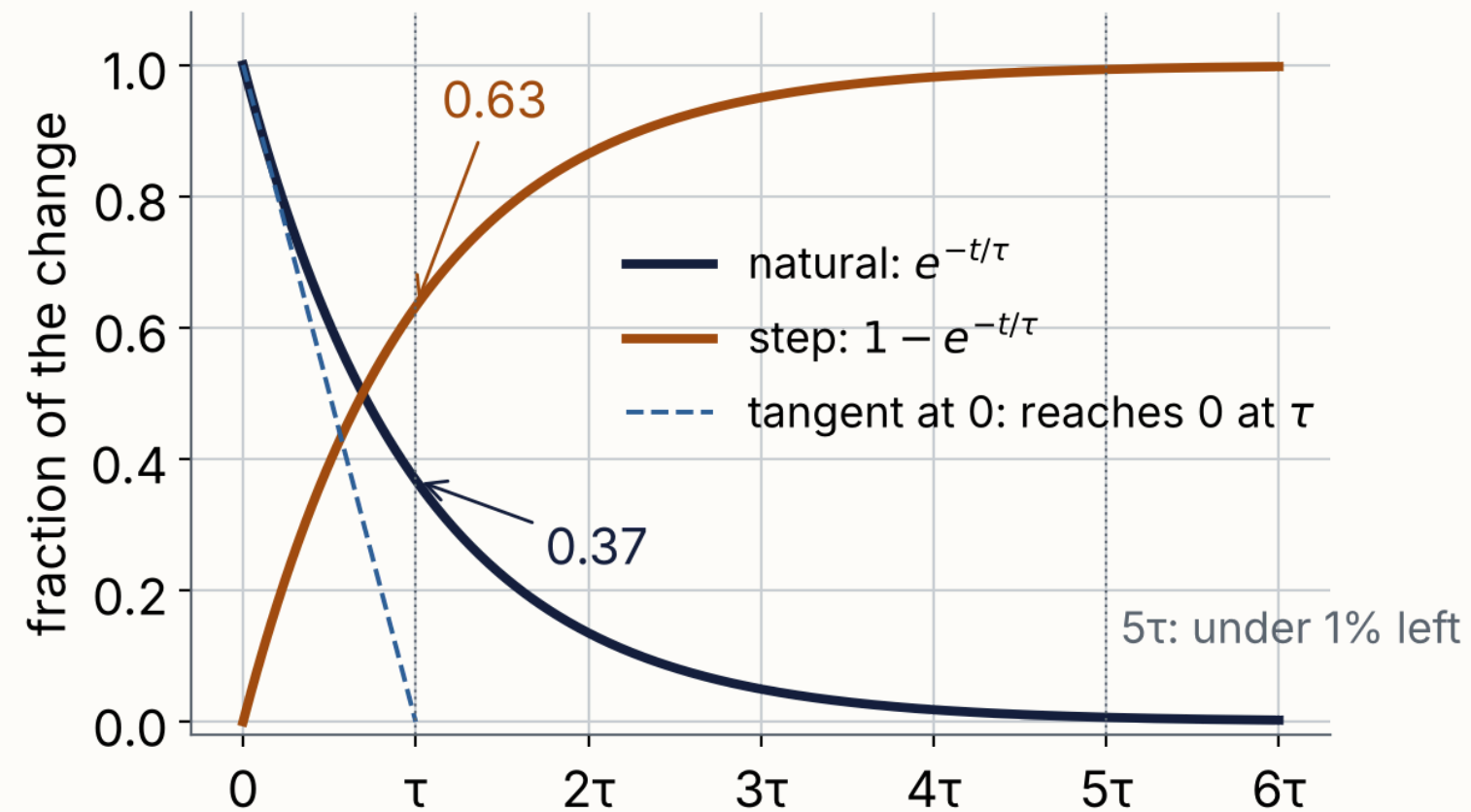
Before $t = 0$, $i_o = 0$ and $v_o = 0$: they jump at $t = 0$, which is why they hold only for $t \geq 0^+$.

Energy: the inductor starts with 400 J; the 10 Ω takes

$$w_{10\Omega} = \int_0^\infty \frac{(160e^{-5t})^2}{10} dt = 256 \text{ J}, \quad \frac{256}{400} = 64\%$$

THE TIME CONSTANT

After one τ , 37% of the change is still to come



After five time constants less than 1% of the change remains: in practice, finished.

The tangent at $t = 0$ reaches the final value after exactly one τ : a quick way to read τ from a measured curve.

$$x(\tau) - x_f = (x_0 - x_f) e^{-1} = 0.37 (x_0 - x_f), \quad e^{-5} = 0.0067$$

NILSSON EXAMPLE 7.2

Parallel inductors: the Chapter 6 trapped energy again

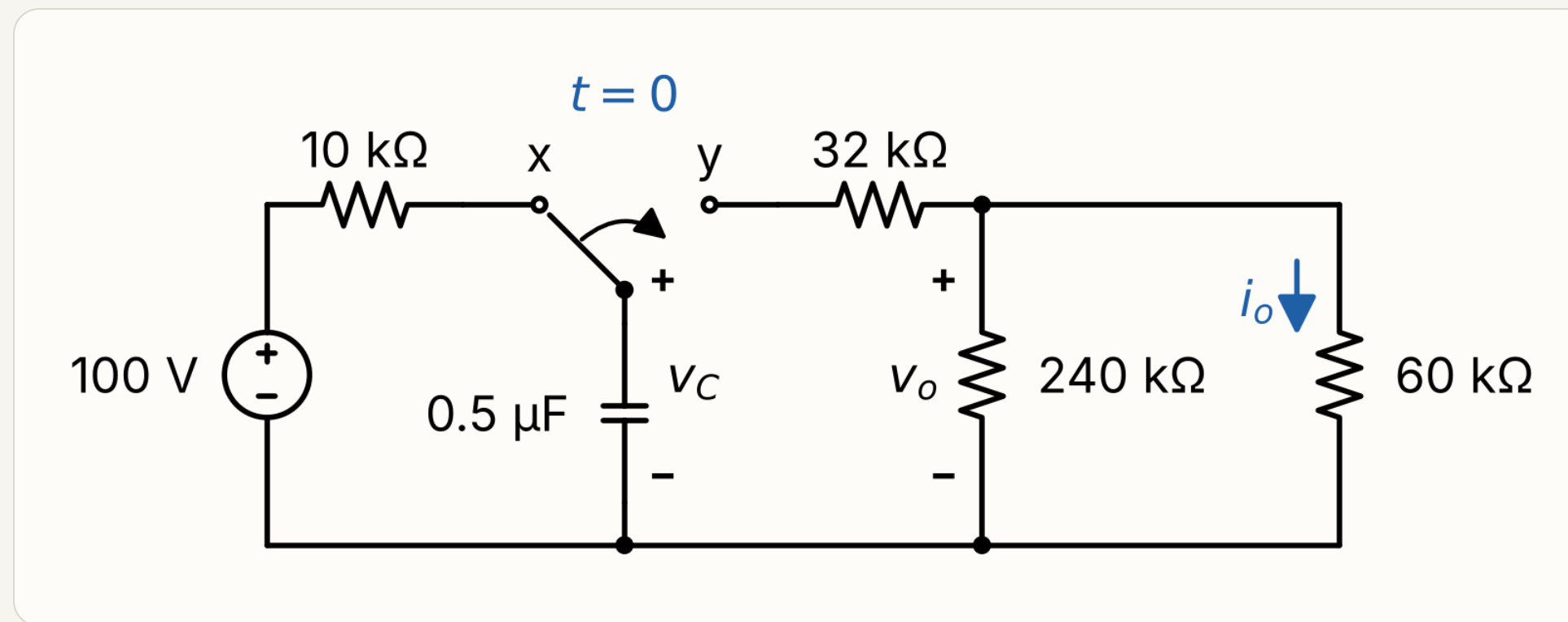
A 5 H (8 A) and a 20 H (4 A) inductor in parallel discharge into a resistive network when a switch opens at $t = 0$.

$$L_{eq} = 4 \text{ H}, \quad I_0 = 12 \text{ A}, \quad R = 40 \parallel (4 + 15 \parallel 10) = 8 \text{ } \Omega, \quad \tau = 0.5 \text{ s}, \quad i = 12e^{-2t} \text{ A}$$

$$w(0) = \frac{1}{2}(5)(8)^2 + \frac{1}{2}(20)(4)^2 = 320 \text{ J}, \quad w(\infty) = 32 \text{ J}, \quad 320 - 32 = 288 \text{ J} = \frac{1}{2}(4)(12)^2$$

The energy of the equivalent inductor is exactly what the resistors receive; 32 J stays trapped with 1.6 A circulating.

The switch moves from x to y at $t = 0$



$$V_0 = 100 \text{ V}, \quad R = 32 \text{ k} + 240 \text{ k} \parallel 60 \text{ k} = 80 \text{ k}\Omega, \quad \tau = (80 \text{ k})(0.5 \mu) = 40 \text{ ms}$$

$$v_C = 100e^{-25t} \text{ V}, \quad v_o = 0.6 v_C = 60e^{-25t} \text{ V}, \quad i_o = \frac{v_o}{60 \text{ k}} = e^{-25t} \text{ mA}$$

NILSSON EXAMPLE 7.4

Series capacitors: the equivalent sets the response

A 5 μF capacitor with $v_1(0) = -4 \text{ V}$ and a 20 μF capacitor with $v_2(0) = 24 \text{ V}$ (Nilsson's references), in series, discharge into 250 $\text{k}\Omega$ when a switch closes at $t = 0$.

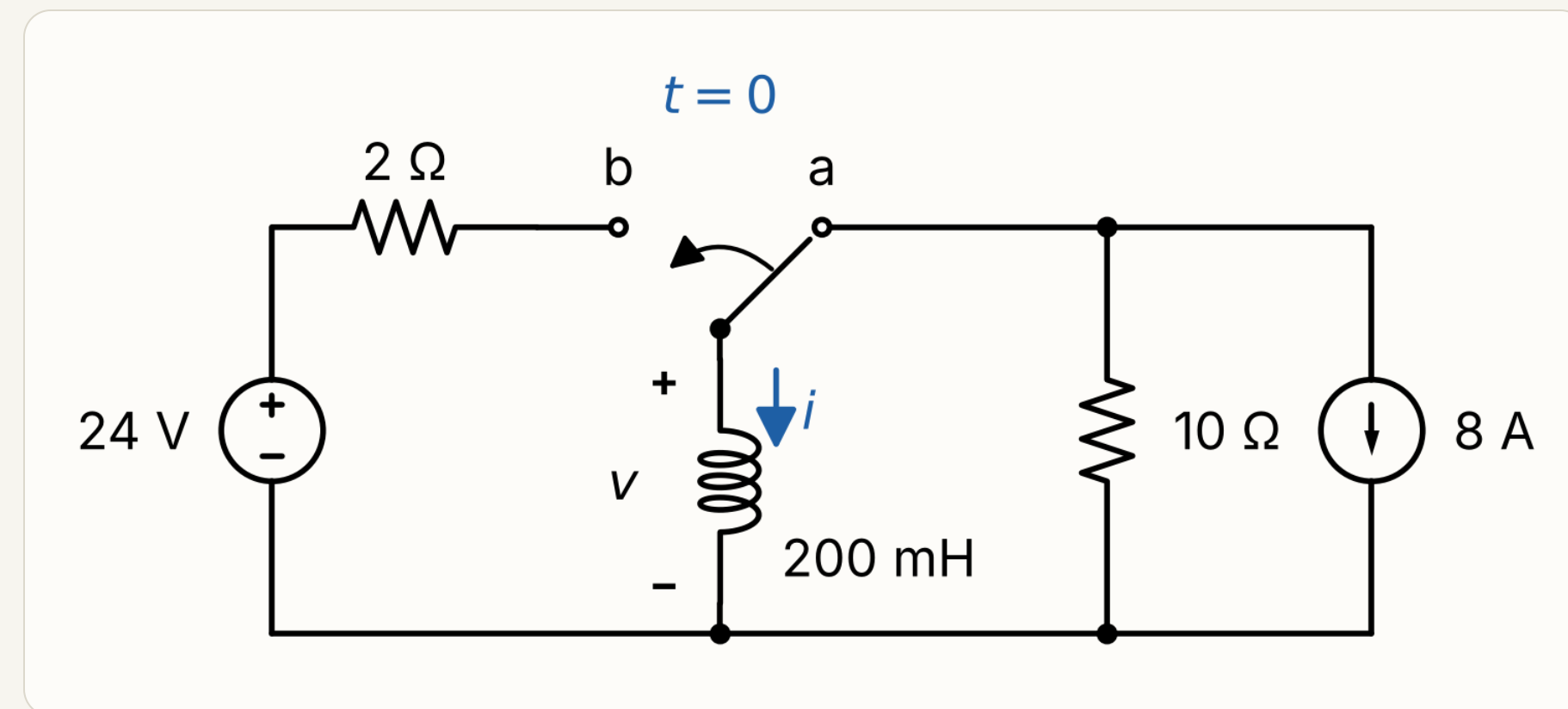
$$C_{eq} = 4 \mu\text{F} \text{ at } 20 \text{ V}, \quad \tau = 1 \text{ s} : \quad v = 20e^{-t} \text{ V}, \quad w_{\text{delivered}} = \frac{1}{2}(4 \mu)(20)^2 = 800 \mu\text{J}$$

The capacitors start with 5800 μJ and end with 5000 μJ trapped ($v_1 \rightarrow -20 \text{ V}$, $v_2 \rightarrow +20 \text{ V}$): only 800 μJ reaches the resistor.

A source drives the inductor current to a new value

$$i(t) = I_f + (I_0 - I_f) e^{-t/\tau}$$

$$v(t) = V_f + (V_0 - V_f) e^{-t/\tau}$$



Make-before-break: the switch touches b before it leaves a, so i never stops.

For $t < 0$ the 8 A source drives current up through the inductor short.

$$I_0 = -8 \text{ A}, \quad I_f = \frac{24}{2} = 12 \text{ A}, \quad \tau = \frac{0.2}{2} = 0.1 \text{ s}, \quad i = 12 - 20e^{-10t} \text{ A}$$

NILSSON EXAMPLE 7.5

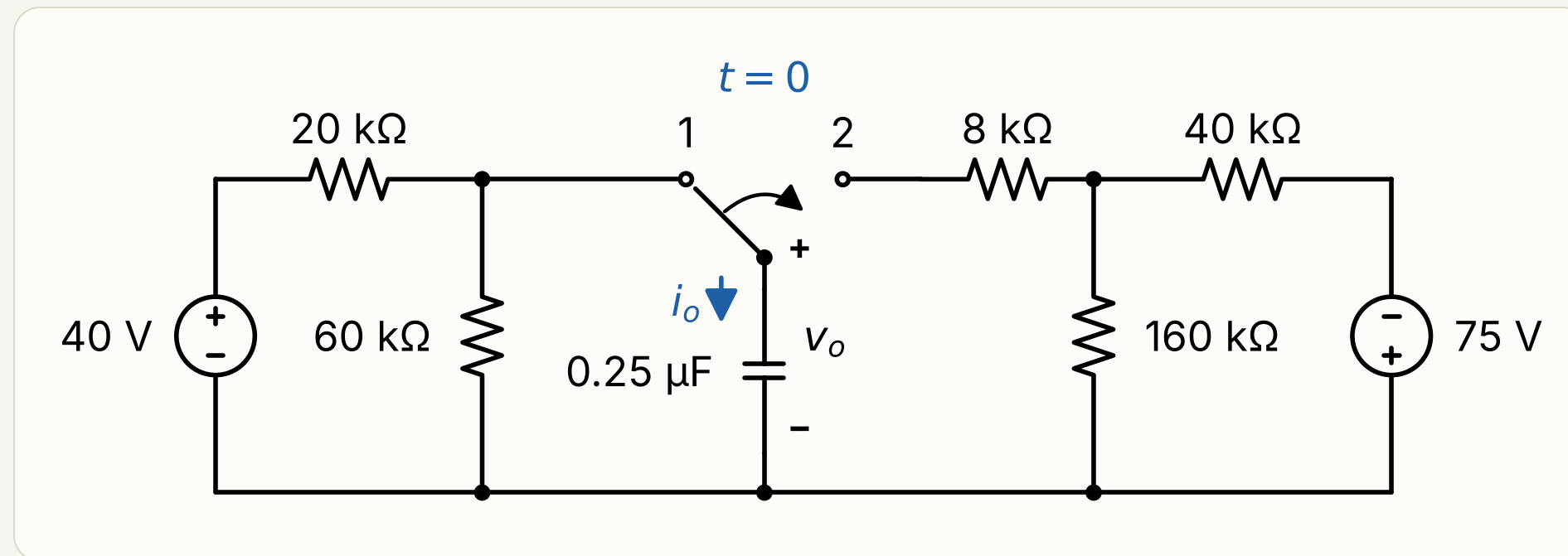
The inductor voltage jumps from 0 to 40 V

$$v = 0.2 \frac{di}{dt} = 40e^{-10t} \text{ V}; \quad 24 = 40e^{-10t} \Rightarrow t = 0.1 \ln \frac{40}{24} = 51.08 \text{ ms}$$

Why 40 V from a 24 V source? At $t = 0^+$ the current is still -8 A, so the 2Ω drops 16 V that adds to the source: $24 + 16 = 40$ V.

Sketch i and v together: i rises from -8 A through zero to 12 A while v decays from 40 V; when $i = 0$ the inductor voltage equals the source voltage.

Replace the rest of the circuit by its Thévenin equivalent



$$V_0 = 40 \frac{60}{80} = 30 \text{ V}, \quad V_{oc} = -75 \frac{160}{200} = -60 \text{ V}, \quad R_{Th} = 8 + 40 \parallel 160 = 40 \text{ k}\Omega, \quad \tau = 10 \text{ ms}$$

$$v_o = -60 + 90e^{-100t} \text{ V}, \quad i_o = C \frac{dv_o}{dt} = -2.25e^{-100t} \text{ mA}$$

THE GENERAL METHOD

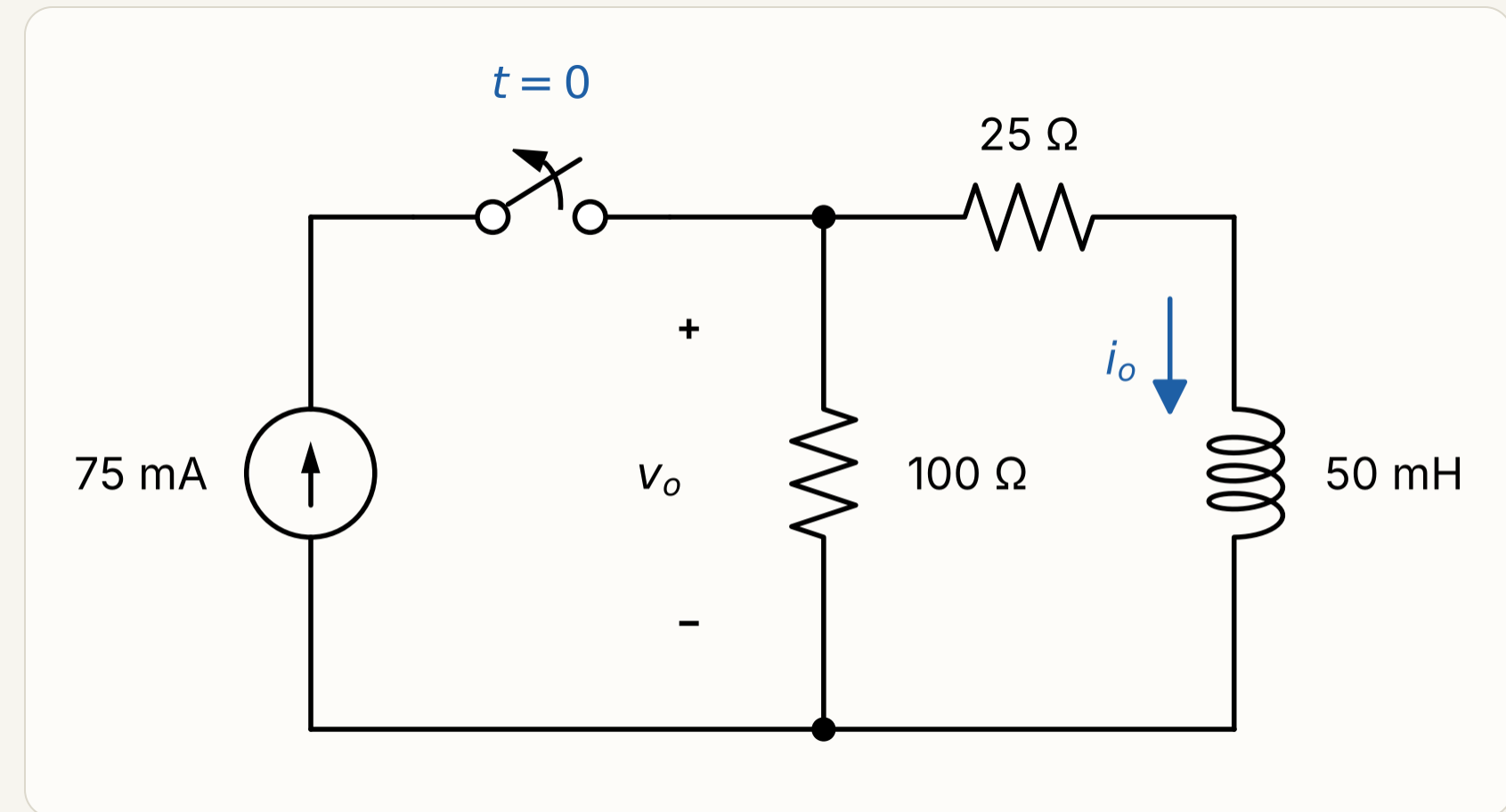
Six steps for every first-order circuit

$$x(t) = x_f + (x_0 - x_f) e^{-t/\tau}, \quad \tau = \frac{L}{R_{Th}} \text{ or } \tau = R_{Th}C$$

- 1** Choose x : the inductor current or the capacitor voltage.
- 2** Find x_0 from the circuit for $t < 0$.
- 3** Find τ from the circuit for $t \geq 0$.
- 4** Find x_f from the dc circuit long after switching.
- 5** Write $x(t)$.
- 6** Find every other quantity from $x(t)$.

NILSSON EXAMPLE 7.7

The general method for a natural response



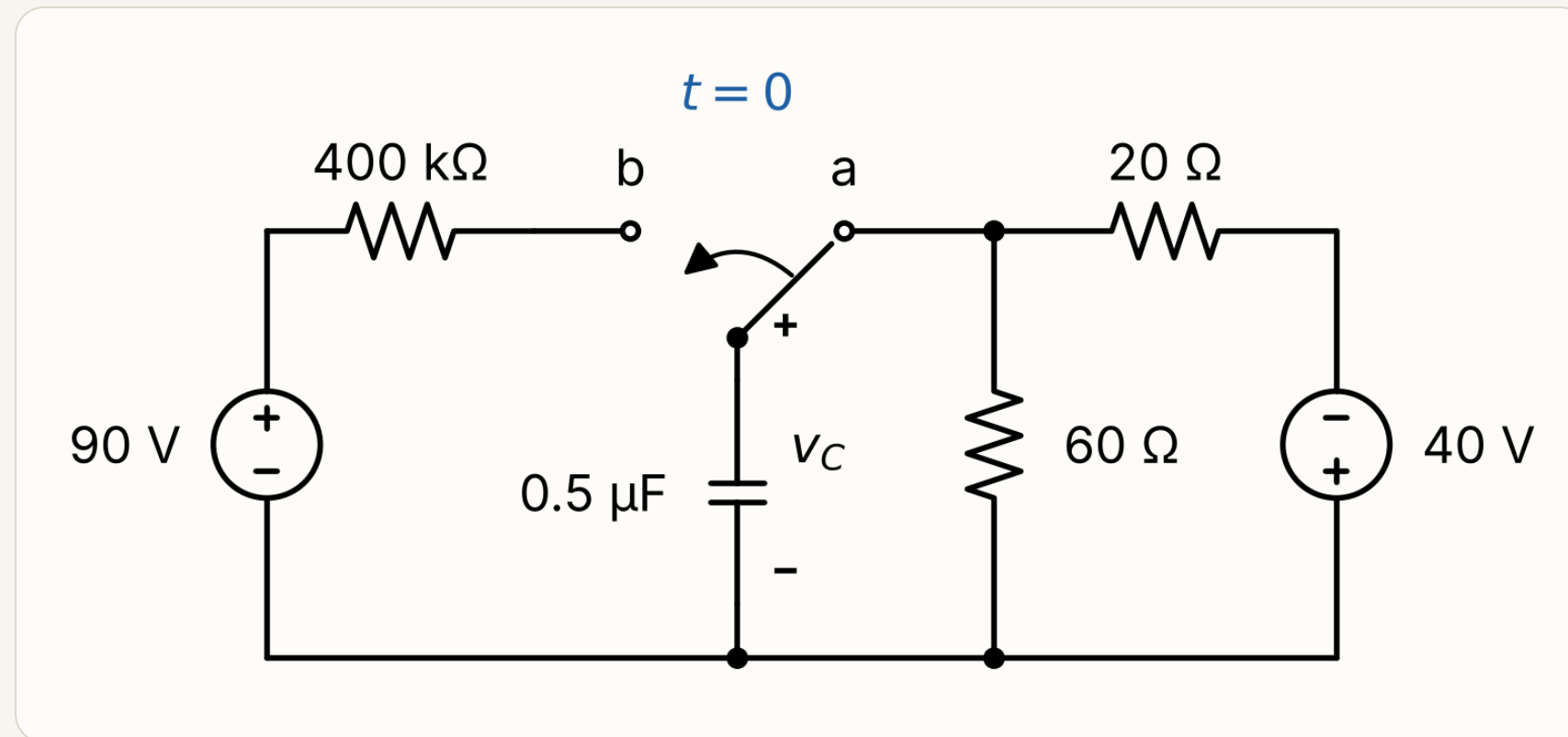
v_o jumps from 1.5 V to -6 V: once the source is gone, the inductor current flows up through the $100\ \Omega$.

$$i_o(0) = 75\text{ m} \frac{100}{125} = 60\text{ mA}, \quad \tau = \frac{0.05}{125} = 0.4\text{ ms}, \quad i_o = 60e^{-2500t}\text{ mA}$$

$$v_o(0^-) = 100(75 - 60)\text{ m} = 1.5\text{ V}, \quad v_o(0^+) = -100(60\text{ m}) = -6\text{ V}$$

NILSSON EXAMPLE 7.8

A step response that starts below zero



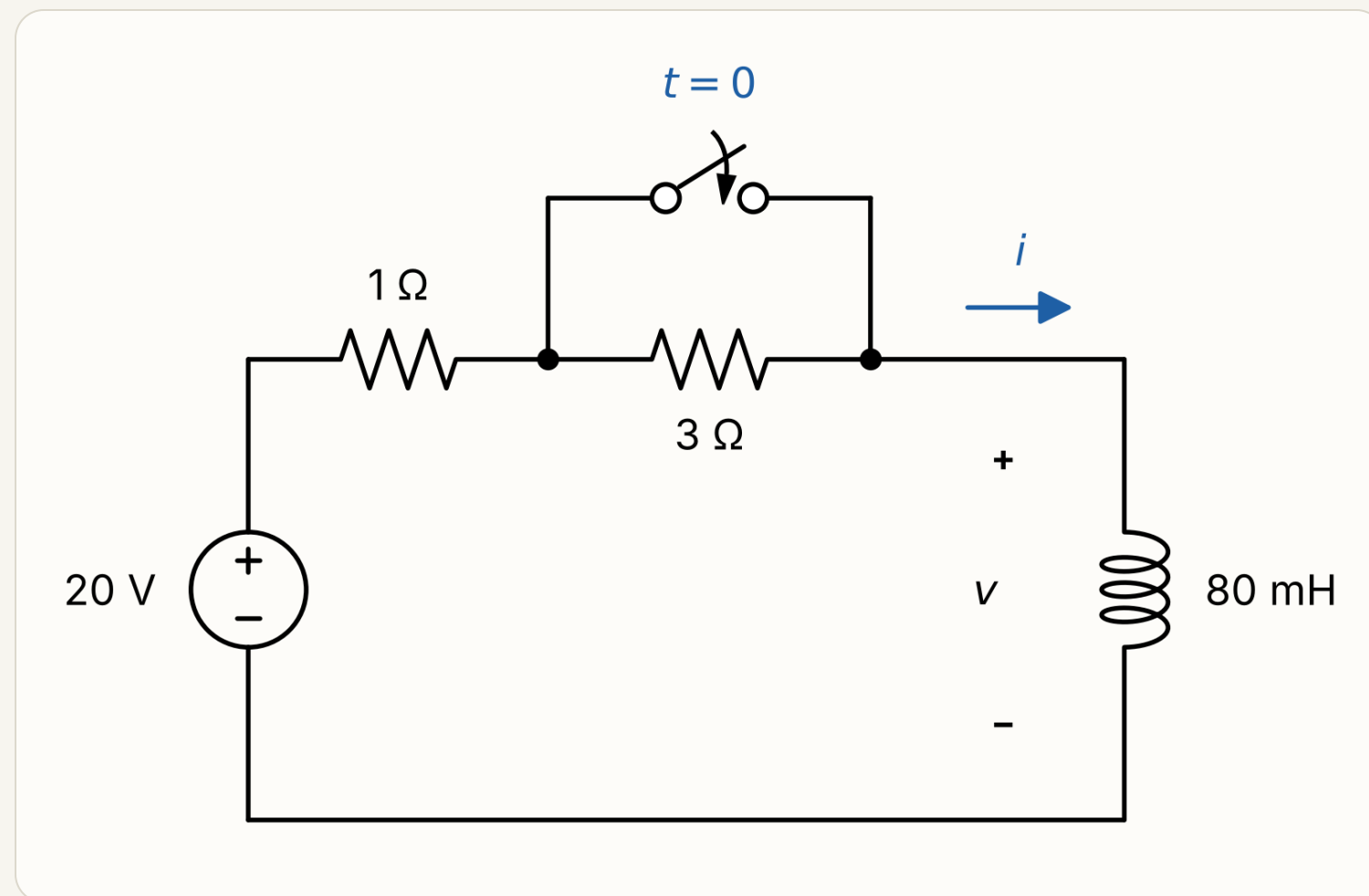
For $t < 0$ the 40 V source's + terminal is at the bottom, so the top of the 60 Ω is negative:

$$v_C(0) = -40 \frac{60}{80} = -30 \text{ V}, \quad V_f = 90 \text{ V}, \quad \tau = (400 \text{ k})(0.5 \mu) = 0.2 \text{ s}, \quad v_C = 90 - 120e^{-5t} \text{ V}$$

$$i = C \frac{dv_C}{dt} = 300e^{-5t} \mu\text{A}; \quad v_C = 0 \text{ at } t = 0.2 \ln \frac{120}{90} = 57.54 \text{ ms}$$

NILSSON EXAMPLE 7.9

A switch that closes and shorts part of the circuit



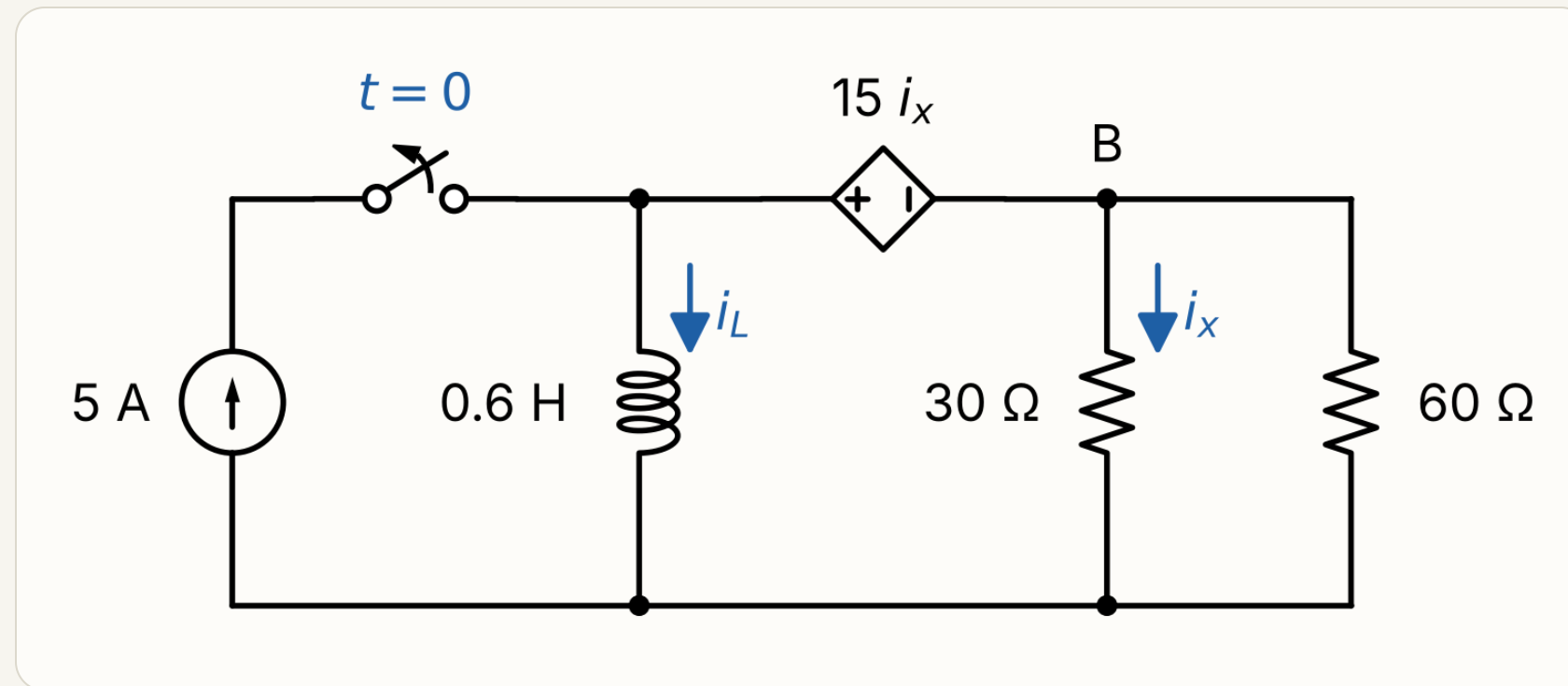
Shorting the $3\ \Omega$ lowers the resistance: the current rises from 5 A to 20 A with $\tau = 80\text{ ms}$.

$$v = L \frac{di}{dt} = 15e^{-12.5t}\text{ V}, \quad t \geq 0^+$$

$$I_0 = \frac{20}{4} = 5\text{ A}, \quad I_f = \frac{20}{1} = 20\text{ A}, \quad \tau = \frac{0.08}{1} = 80\text{ ms}, \quad i = 20 - 15e^{-12.5t}\text{ A}$$

A DEPENDENT SOURCE

τ needs the Thévenin resistance: use a test source



Switch the independent source off, keep the dependent one, and replace the inductor by a test source i_T that drives current into the network at the inductor's top terminal (voltage v_T):

$$v_B = i_T (30 \parallel 60) = 20 i_T, \quad i_x = \frac{v_B}{30} = \frac{2}{3} i_T, \quad v_T = v_B + 15 i_x = 30 i_T \Rightarrow R_{Th} = 30 \Omega$$

$$i_L(0) = 5 \text{ A}, \quad \tau = \frac{0.6}{30} = 20 \text{ ms}, \quad i_L = 5e^{-50t} \text{ A}$$

CHECKING AN AI ANSWER

An AI solved Example 7.8 like this. What is wrong?

$$v_C(0) = +30 \text{ V} \Rightarrow v_C = 90 + (30 - 90)e^{-5t} = 90 - 60e^{-5t} \text{ V}$$

Correct:

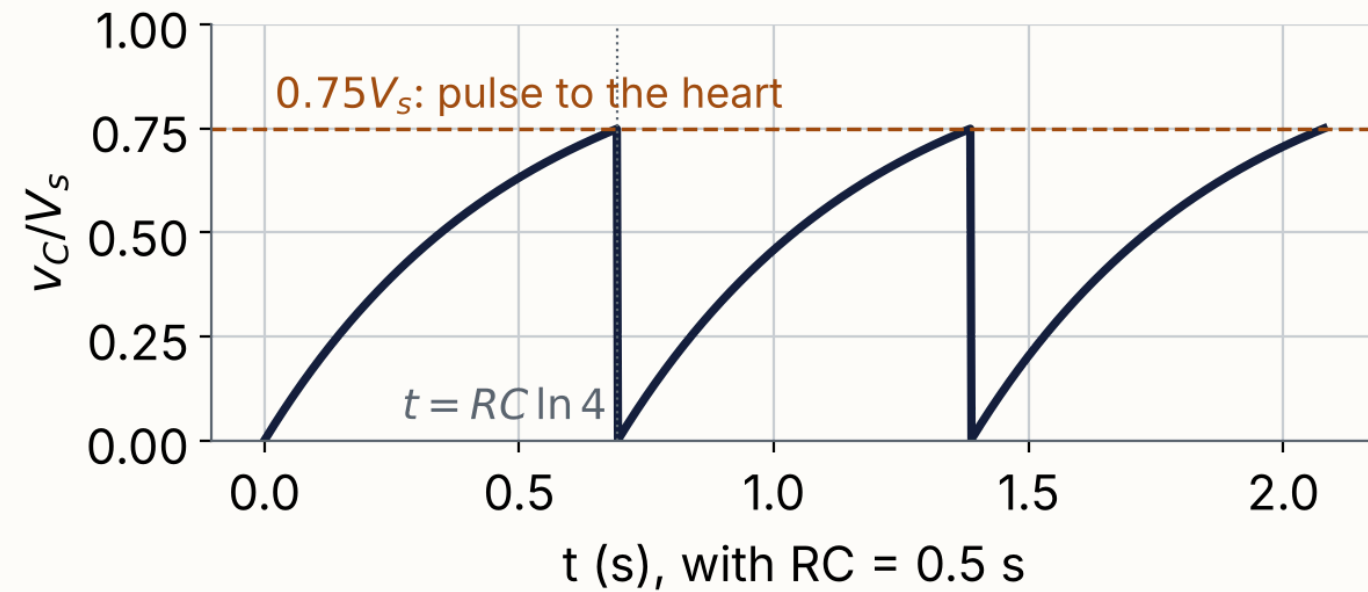
$$v_C(0) = -30 \text{ V} \Rightarrow v_C = 90 - 120e^{-5t} \text{ V}$$

It read the divider voltage but not its polarity: the 40 V source's + terminal is at the bottom, so $v_C(0) = -30 \text{ V}$. Putting $t = 0$ into its answer just returns its own 30 V; only the circuit for $t < 0$ catches the slip.

In a 2025 study of Gemini 2.5 Pro on undergraduate circuit problems, misread source polarities caused 6 of its 17 wrong answers, and misread current directions 5.

PRACTICAL PERSPECTIVE: AN ARTIFICIAL PACEMAKER

Each heartbeat is one RC charging curve



Back to slide 2: doubling R doubles RC , so each beat takes twice as long and the rate halves.

$$v_C = V_s \left(1 - e^{-t/RC}\right) = 0.75 V_s \Rightarrow t = RC \ln 4, \quad H = \frac{60}{RC \ln 4} \text{ beats per minute}$$

$$RC = (250 \text{ k}\Omega)(2 \text{ }\mu\text{F}) = 0.5 \text{ s} : H = 86.6 \text{ beats per minute;} \quad R \text{ doubled} : H = 43.3 \text{ beats per minute}$$

CHECK EVERY ANSWER

Three checks, no second solution needed

$x(0^+) = x(0^-)$ guaranteed only for i_L and v_C , $x(\infty)$ from the dc circuit, $\tau > 0$

Put $t = 0$ in

Your $x(t)$ must return the value from the circuit before switching, with its sign.

Let t grow

$x(\infty)$ must match the dc circuit long after switching.

Look at τ

It comes from the circuit after switching, with the independent sources off, and it is positive.

WHAT COMES NEXT

Practise openly, then show it on your own

In class

Booklet Problems 7.1 to 7.12.
Problem 7.13 is homework.

At home

The 16-item practice sheet. AI
tools allowed; not marked.

Week 14

15-minute quiz, no devices.
Every question is a twin of a
practice-sheet question.

Weeks 14–15: catch-up and revision. The final exam covers Chapters 4 to 7.

CHECKLIST

You can do these without help

- Find the initial inductor current or capacitor voltage from the circuit before switching
- Find the final value from the circuit long after switching
- Find τ from the resistance the element sees, with a test source when there is a dependent source
- Write the response with the general formula and find other currents and voltages from it
- Say which quantities can jump at $t = 0$ and find their values at $t = 0^+$
- Work backwards from a measured response to τ , R , L or C